



NCI Protocol #: 10323

Patient ID:

Date:

1 of 11

Patient Name:

Date Collected:

Biopsy Site: **Colon**

Sample ID:

Telephone:

Tumor Content (%): **70%**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Colorectal adenocarcinoma**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Relevant Colorectal Cancer Findings

Gene	Finding
BRAF	None detected
KRAS	KRAS G12S
NRAS	None detected
NTRK1	None detected
NTRK2	None detected
NTRK3	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	KRAS G12S Prognostic significance: None Diagnostic significance: None	None	None	45
III	PDGFRB V633I, PDGFRB D111N Prognostic significance: None Diagnostic significance: None	None	None	3

Public data sources included in relevant therapies: FDA1, NCCN

Public data sources included in prognostic and diagnostic significance: NCCN

Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

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Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.03(006). The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

Relevant Biomarkers (continued)

 Alerts informed by public data sources:  Contraindicated,  Resistance

KRAS G12S  cetuximab¹, panitumumab¹

Public data sources included in alerts: FDA¹, NCCN

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
PDGFRB	p.(V633I)	c.1897G>A	.	chr5:149504305	85.37%	NM_002609.3	missense
PDGFRB	p.(D111N)	c.331G>A	.	chr5:149515151	85.34%	NM_002609.3	missense
KRAS	p.(G12S)	c.34G>A	COSM517	chr12:25398285	48.47%	NM_033360.3	missense

Comments:

N/A

Signed By: _____

Biomarker Descriptions

KRAS (KRAS proto-oncogene, GTPase)

Background: The KRAS proto-oncogene encodes a GTPase that functions in signal transduction and is a member of the RAS superfamily which also includes NRAS and HRAS. RAS proteins mediate the transmission of growth signals from the cell surface to the nucleus via the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways, which regulate cell division, differentiation, and survival^{1,2,3}.

Alterations and prevalence: Recurrent mutations in RAS oncogenes cause constitutive activation and are found in 20-30% of cancers. KRAS mutations are observed in up to 10-20% of uterine cancer, 30-35% of lung adenocarcinoma and colorectal cancer, and about 60% of pancreatic cancer⁴. The majority of KRAS mutations consist of point mutations occurring at G12, G13, and Q61^{4,5,6}. Mutations at A59, K117, and A146 have also been observed but are less frequent^{7,8}.

Potential relevance: The KRAS inhibitor, sotorasib⁹, is approved (2021) for the treatment of adult patients with KRAS G12C-mutated locally advanced or metastatic non-small cell lung cancer (NSCLC). The FDA has granted breakthrough therapy designation (2021) to the small molecule inhibitor, adagrasib, for KRAS G12C positive in non-small cell lung cancer following prior systemic therapy¹⁰. The small molecular inhibitor, RO-5126766, was also granted breakthrough designation (2021) alone for KRAS G12V mutant non-small cell

Biomarker Descriptions (continued)

lung cancer or in combination with defactinib, for KRAS mutant endometrial carcinoma and KRAS G12V mutant non-small cell lung cancer¹¹. Additionally, onvansertib¹² was granted fast track designation (2020) for second-line treatment of patients with KRAS-mutated metastatic colorectal cancer (mCRC). The EGFR antagonists, cetuximab¹³ and panitumumab¹⁴, are contraindicated for treatment of colorectal cancer patients with KRAS mutations in exon 2 (codons 12 and 13), exon 3 (codons 59 and 61), and exon 4 (codons 117 and 146)⁸. Additionally, KRAS mutations are associated with poor prognosis in NSCLC¹⁵.

PDGFRB (platelet derived growth factor receptor beta)

Background: The PDGFRB gene encodes the platelet derived growth factor receptor beta, a member of the PDGF receptor type III receptor tyrosine kinase family, which includes PDGFRA, CSF1R, FLT1, FLT3, FLT4, KDR, and KIT^{16,17}. PDGFRB is a receptor for platelet derived growth factors, which are mitogens for cells of mesenchymal origin¹⁸. PDGFRB may function as a homodimer or heterodimer with PDGFRA depending on the ligand¹⁹. Ligand binding to PDGFRB results in receptor dimerization, kinase activation, and stimulation of several downstream pathways promoting cell proliferation and survival²⁰. PDGFRB is the target of recurrent chromosomal translocations in myeloid malignancies including chronic myelomonocytic leukemia (CMML)²¹. Such fusions have been reported to result in constitutive pathway activation suggesting an oncogenic role for PDGFRB in cancer²². PDGFRB is one of many genes associated with deletion 5q, the most common cytogenetic abnormality that contributes to a rare form of myelodysplastic syndrome (MDS)^{23,24}. Additionally, germline mutations in PDGFRB are causal of familial infantile myofibromatosis, a common benign tumor observed in infancy and childhood²⁵.

Alterations and prevalence: The t(5;12)(q33;p13) translocation, which results in PDGFRB-ETV6 fusion, is recurrent in CMML^{21,26}. Other PDGFRB rearrangements are also observed, but are less common²³. Mutations in PDGFRB are also reported in solid tumors including approximately 7% of melanoma and uterine cancer as well as approximately 3% of lung, stomach, and colorectal cancers⁷.

Potential relevance: The small molecule kinase inhibitor, imatinib²⁷ (2001) is approved for PDGFRB fusion positive myelodysplastic/myeloproliferative diseases (MDS/MPD). In one study, an acquired PDGFRB missense mutation, C843G, was reported to confer resistance to tyrosine kinase inhibitors including imatinib, dasatinib, and ponatinib in a patient with Philadelphia chromosome-like acute lymphoblastic leukemia (Ph-like ALL) possessing a novel AGGF1-PDGFRB fusion²⁸.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

KRAS G12S

Relevant Therapy	FDA	NCCN	Clinical Trials*
apatinib, chemotherapy	×	×	● (IV)
bevacizumab, chemotherapy	×	×	● (III)
atezolizumab + cobimetinib	×	×	● (II)
cabozantinib, nivolumab	×	×	● (II)
cetuximab, chemotherapy	×	×	● (II)
nivolumab, chemotherapy, bevacizumab	×	×	● (II)
panitumumab, trametinib	×	×	● (II)
pembrolizumab	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

KRAS G12S (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
selinexor, pembrolizumab	×	×	● (II)
ASN007, palbociclib	×	×	● (I/II)
ASTX029	×	×	● (I/II)
cobimetinib, atezolizumab, antimalarial	×	×	● (I/II)
eltanexor	×	×	● (I/II)
HH-2710	×	×	● (I/II)
mirdametinib, lifirafenib	×	×	● (I/II)
navitoclax, trametinib	×	×	● (I/II)
neratinib, valproic acid	×	×	● (I/II)
OMO-103	×	×	● (I/II)
onvansertib, chemotherapy, bevacizumab	×	×	● (I/II)
regorafenib, chemotherapy	×	×	● (I/II)
RMC-4630, pembrolizumab	×	×	● (I/II)
SX-682, nivolumab	×	×	● (I/II)
zotatifin	×	×	● (I/II)
AZD-0364	×	×	● (I)
BBP-398	×	×	● (I)
BI-1701963, BI-3011441	×	×	● (I)
BI-3011441	×	×	● (I)
CDK 4/6 inhibitor, MEK inhibitor	×	×	● (I)
chemotherapy, binimetinib	×	×	● (I)
cobimetinib, belvarafenib	×	×	● (I)
COM701, nivolumab	×	×	● (I)
DCC-3116, trametinib	×	×	● (I)
GH35	×	×	● (I)
HBI 2376	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

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Relevant Therapy Summary (continued)

☒ In this cancer type

☐ In other cancer type

☒ In this cancer type and other cancer types

☒ No evidence

KRAS G12S (continued)			
Relevant Therapy	FDA	NCCN	Clinical Trials*
JAB-3312	×	×	● (I)
JSI-1187	×	×	● (I)
neratinib, trametinib	×	×	● (I)
palbociclib, binimetinib	×	×	● (I)
PF-07284892, binimetinib	×	×	● (I)
RMC-4630	×	×	● (I)
siremadlin, trametinib	×	×	● (I)
TNO-155, ribociclib	×	×	● (I)
trametinib, ruxolitinib	×	×	● (I)
ulixertinib, antimalarial	×	×	● (I)

PDGFRB V633I, PDGFRB D111N			
Relevant Therapy	FDA	NCCN	Clinical Trials*
dasatinib, sunitinib	×	×	● (II)
nilotinib, pazopanib	×	×	● (II)
regorafenib	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

KRAS G12S		
NCT ID	Title	Phase
NCT04835714	A Phase I Open-Label Dose Escalation Trial Of BI 1701963 As Monotherapy And In Combination With BI 3011441 In Patients With KRAS Mutated Advanced Or Metastatic Solid Tumours	I
NCT04045496	A Phase I, Multi-Center, Open-Label Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Preliminary Evidence of Antitumor Activity of JAB-3312 in Adult Patients With Advanced Solid Tumors	I
NCT04000529	A Phase Ib, Open-label, Multi-center Study to Characterize the Safety, Tolerability, and Preliminary Efficacy of TNO155 in Combination With Spartalizumab or Ribociclib in Selected Malignancies	I

Clinical Trials Summary (continued)

KRAS G12S (continued)

NCT ID	Title	Phase
NCT02885753	Systemic Oxaliplatin or Intra-arterial Chemotherapy Combined With LV5FU2 and an Target Therapy in First Line Treatment of Metastatic Colorectal Cancer Restricted to the Liver	III
NCT03087071	A Phase II Enrichment Study of Panitumumab as a Single Agent or in Combination With Trametinib in Anti-EGFR-Refractory Stage IV Colorectal Cancer Patients	II
NCT04214418	Phase I/II Open-label Study of Combination Therapy With The MEK Inhibitor, Cobimetinib, Immune Checkpoint Blockade, Atezolizumab, And The AUTophagy Inhibitor, Hydroxychloroquine In KRAS-mutated Advanced Malignancies	I/II
NCT03829410	A Phase Ib/II Study of Onvansertib (PCM-075) in Combination With FOLFIRI and Bevacizumab for Second Line Treatment of Metastatic Colorectal Cancer in Patients With a KRAS Mutation.	I/II
No NCT ID	Exploratory Study Of Apatinib For Advanced Colorectal Cancer	IV
NCT04189055	Cetuximab as Salvage Therapy in Patients with Neo Wild-type RAS/RAF Metastatic Colorectal Cancer with Liver Metastases. A Proof-of-concept Study	II
NCT03519412	Pembrolizumab in MMR-Proficient Metastatic Colorectal Cancer Pharmacologically Primed to Trigger Dynamic Hypermutation Status	II
NCT05039177	A Phase 1b/2 Study of Agents Targeting the Mitogen-Activated Protein Kinase Pathway in Patients With Advanced Gastrointestinal Malignancies (HERKULES-3)	I/II
NCT04418661	A Phase 1/2, Open-label, Multicenter, Dose Escalation and Dose Expansion Study of SAR442720 in Combination With Pembrolizumab in Patients With Advanced Malignancies	I/II
NCT04615312	Phase I Clinical Study on the Safety and Tolerability of a CDK4 / 6 Inhibitor and a MEK Inhibitor in the Treatment of Metastatic Digestive System Tumors	I
NCT02613650	A Phase Ib Trial of a Combination of mFOLFIRI With MEK162 in Patients With Advanced RAS (HRAS, NRAS, or KRAS) Positive Metastatic Colorectal Cancers	I
NCT03667716	A Phase Ia/Ib Study of COM701 as Monotherapy and In Combination With an Anti-PD-1 Antibody in Subjects With Advanced Solid Tumors	I
NCT04892017	A Phase I, First-in-Human Study of DCC-3116 as a Single Agent and in Combination With Trametinib in Patients With Advanced or Metastatic Solid Tumors With RAS or RAF Mutations.	I
NCT04870034	Perioperative Analysis of Binimetinib and Palbociclib in RAS-Driven Tumors	I
NCT03714958	A Single-center, Phase I Dose Escalation Study of Trametinib Combined With HDM201 in Patients With RAS/RAF Mutant and TP53 Wild-type Advanced/Metastatic Colorectal Cancer.	I
NCT04145297	A Phase I Trial of Ulixertinib (BVD-523) and Hydroxychloroquine in Patients With Advanced MAPK-Mutated Gastrointestinal Adenocarcinomas	I
NCT02649790	A Phase I/II Open-Label Study of the Safety, Tolerability and Efficacy of the Selective Inhibitor of Nuclear Export (SINE) Compound Eltanexor (KPT-8602) in Patients With Newly Diagnosed and Relapsed/Refractory Cancer Indications	I/II
NCT04781270	mFOLFOXIRI Plus Bevacizumab Versus mFOLFOX6 Plus Bevacizumab for the First Line Treatment of RAS Mutant Unresectable Colorectal Liver-limited Metastases	III

Clinical Trials Summary (continued)

KRAS G12S (continued)

NCT ID	Title	Phase
NCT04072198	Phase II Study on NIVolumab in Combination With FOLFOXIRI/Bevacizumab in First Line Chemotherapy of Advanced COloRectal Cancer RASm/BRAFm Patients.	II
NCT04854434	A Phase II Open-Label Multicenter Study to Evaluate the Safety and Efficacy of Selinexor With or Without Pembrolizumab Versus Standard of Care in Previously Treated Metastatic Colorectal Cancer With RAS Mutations	II
NCT03905148	A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors	I/II
NCT04808362	A Phase I/II Study to Evaluate the Safety, Pharmacokinetics, and Anti-Tumour Activity of the MYC Inhibitor OMO-103 Administered Intravenously in Patients With Advanced Solid Tumours	I/II
NCT03828799	Safety, Tolerability and Efficacy of Regorafenib in Combination with FOLFIRINOX in Patients with RAS-mutated Metastatic Colorectal Cancer: a Dose-escalation, Phase I/II Trial	I/II
NCT03284502	A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT04963283	A Phase II Study of Cabozantinib and Nivolumab in Refractory Metastatic Microsatellite Stable (MSS) Colorectal Cancer	II
NCT02079740	An Open Label, Two-Part, Phase Ib/II Study to Investigate the Safety, Pharmacokinetics, Pharmacodynamics, and Clinical Activity of the MEK Inhibitor Trametinib and the BCL2-Family Inhibitor Navitoclax (ABT-263) in Combination in Subjects With KRAS or NRAS Mutation-Positive Advanced Solid Tumors	I/II
NCT04092673	A Phase 1-2 Dose-Escalation and Cohort-Expansion Study of Intravenous Zotatfin (eFT226) in Subjects With Selected Advanced Solid Tumor Malignancies	I/II
NCT04742556	A Phase I Open-label Trial of BI 3011441 in Japanese Patients With NRAS/KRAS Mutation Positive Advanced, Unresectable or Metastatic Refractory Solid Tumours	I
NCT03919292	Phase 1/2 Study of Neratinib and Divalproex Sodium (Valproate) in Advanced Solid Tumors, With an Expansion Cohort in Ras-Mutated Cancers	I/II
NCT04599140	Phase Ib/II Trial of SX-682 in Combination With Nivolumab for Refractory RAS Mutated (RAS) Microsatellite Stable (MSS) Metastatic Colorectal Cancer (mCRC) (STOPTRAFFIC-1)	I/II
NCT05010694	A Phase I Study Evaluating the Safety, Tolerability, Pharmacokinetic Characteristics, and Primary Antitumor Activity of GH35 in Patients With Advanced Solid Tumors With KRAS Mutation	I
NCT05163028	A Phase I, Open-Label, Dose Escalation of HBI-2376 in Patients With Advanced Malignant Solid Tumors Harboring KRAS or EGFR Mutations	I
NCT04418167	A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations	I
NCT03065387	Phase I Study of the Pan-ERBB Inhibitor Neratinib Given in Combination With Everolimus, Palbociclib, or Trametinib in Advanced Cancer Subjects With EGFR Mutation/Amplification, HER2 Mutation/Amplification, or HER3/4 Mutation or KRAS Mutation	I

Clinical Trials Summary (continued)

KRAS G12S (continued)

NCT ID	Title	Phase
NCT04800822	A Phase I Open-Label, Multi-Center, Dose Escalation and Dose Expansion Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Preliminary Evidence of Antitumor Activity of PF-07284892 (ARRY-558) as a Single Agent and in Combination Therapy in Participants With Advanced Solid Tumors	I
NCT04303403	Phase Ib Study Evaluating Safety and Tolerability of Combination Trametinib and Ruxolitinib in Patients with Advanced RAS Mutant Colorectal Cancer and Pancreatic Adenocarcinoma	I
NCT04198818	A First-in-Human, Open Label, Phase I/II Study to Evaluate the Safety, Tolerability and Pharmacokinetics of HH2710 in Patients With Advanced Tumors	I/II
NCT03634982	A Phase I, Open-Label, Multicenter, Dose-Escalation Study of RMC-4630 Monotherapy in Adult Participants with Relapsed/Refractory Solid Tumors	I
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II
NCT03520075	A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors	I/II
NCT04305249	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies	I
NCT04528836	A Phase I/IB First-in-Human Study of the SHP2 Inhibitor BBP-398 (Formerly Known as IACS-15509) in Patients With Advanced Solid Tumors	I

PDGFRB V633I, PDGFRB D111N

NCT ID	Title	Phase
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients with Progressive Locally-advanced or Metastatic Solid Tumors.	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent OncoPrint Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OncoPrint Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trial, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSP02, RSP03, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

References

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